Using Zero and Negative Exponents
Answer Key
Grade 8 / Pre-Algebra

Quick Answers

1. 1
2. 2
3. $1 \cdot \frac{1}{8}$
4. $1 \cdot \frac{1}{25}$
5. 1
6. 1
7. $1\frac{1}{y^6}$
8. $1\frac{1}{x^5}$
9. $1 \cdot \frac{1}{10000}$
10. $12m^3$
11. $>$
12. (a) the correct value $= 1 \cdot \frac{1}{9}$, —

What is verified

11 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra
8.EE.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12*
8.EE.A.4 – *problem 9*
Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

12. *If they got -9: read the negative exponent as a minus sign on the answer*

1. Evaluate 7^0.
Solution: Any non-zero base to the power zero is one. Dividing that base's first power by itself is the proof: the quotient is one, and the quotient law writes the same division with exponent zero. $\boxed{1}$

2. Evaluate $4^0 + 9^0$.
Solution: Each term is a non-zero base to the power zero, so each is one, and one plus one is the sum. The bases play no part. $\boxed{2}$

3. Evaluate as an exact fraction: 2^{-3}.

Solution: A negative exponent means the reciprocal, so this is one over the third power of the base, and that power is eight. $\boxed{\frac{1}{8}}$

4. Evaluate as an exact fraction: 5^{-2}.

Solution: Flip first, then evaluate: one over the second power of the base, and that power is twenty-five. $\boxed{\frac{1}{25}}$

5. Simplify $x^3 \cdot x^{-3}$.

Solution: Same base, so the product law adds the exponents: $3 + (-3) = 0$, giving the base to the power zero. $\boxed{1}$

6. Simplify $(2n)^0$, for n not zero.

Solution: The zero exponent sits on the whole bracket, so the entire quantity inside — the number factor and the variable together — is raised to the power zero. $\boxed{1}$

7. Rewrite with a positive exponent: y^{-6}.

Solution: The negative exponent moves the base across the fraction bar and its sign flips. $\boxed{\dfrac{1}{y^6}}$

8. Simplify with a positive exponent: $\dfrac{x^2}{x^7}$.

Solution: The quotient law subtracts: $2-7 = -5$, giving x^{-5}. There are more factors on the bottom than the top, so the leftovers stay in the denominator. $\boxed{\dfrac{1}{x^5}}$

9. Evaluate as an exact fraction: 10^{-4}.

Solution: One over the fourth power of ten. Multiplying ten by itself four times gives ten thousand, so this is one ten-thousandth — the same factor that a negative power of ten contributes in scientific notation. $\boxed{\frac{1}{10000}}$

10. Simplify with a positive exponent: $(3m^{-2})(4m^5)$.

Solution: Group the number factors and the powers: $3 \cdot 4$ for the numbers, and the product law for the powers, adding $-2 + 5$. The exponent comes out positive, so nothing needs moving across the bar. $\boxed{12m^3}$

11. Write $<$, $>$, or $=$ between 6^0 and 6^{-1}.

Solution: The zero power is one; the exponent -1 gives the reciprocal of the base, which is a proper fraction. One is larger than a proper fraction, so the zero power is the bigger quantity — and notice both are positive, even though one exponent is negative. $\boxed{>}$

12. **Challenge.** Dara writes $3^{-2} = -9$. (a) Correct value? (b) What does the negative exponent do?

Solution (a): Reciprocal first: one over the second power of the base, and that power is nine. $\boxed{\frac{1}{9}}$

Solution (b) — instructor-judged. Dara read the exponent's minus sign as a minus sign on the answer. A negative exponent means reciprocal: the base moves to the denominator, and the value stays positive — it simply becomes small. The pattern makes it plain: each time the exponent drops by one the value is divided by the base, so a positive base can never cross into negative values. Full credit requires that reciprocal statement, not just the right fraction.